EVERY MAP FROM A FIXED FAMILY OF SHEAVES FACTORS THROUGH AN EARLIER STAGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the counting step and the two cited inputs are all named, and no step is left to the reader.

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The transfinite construction of the injective replacements. Same chapter, the construction of the sheaves attached to an ordinal and the functor used in it. Supplies the stages through which the factorisation is asserted and the inclusions between them.
- Cardinals of maps out of a set. Sets, Lemma sets-lemma-map-from-set-lifts. Supplies the lifting statement that bounds the set of maps needing to be killed at each stage.
- Ordinals of arbitrarily large cofinality. Sets, Proposition sets-proposition-exist-ordinals-large-cofinality. Supplies the existence of an ordinal of cofinality larger than the relevant cardinal, which makes the bound effective.

Source reference. Original lemma 0GS0: sources/stacks-sites/pinned-original-source.tex, source lines 3921–3921. Navigation pointer invoked at proof line 984; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05N2: sources/stacks-sets/pinned-original-source.tex, source lines 410–410. Navigation pointer invoked at proof line 989; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original proposition 05N3: sources/stacks-sets/pinned-original-source.tex, source lines 434–434. Navigation pointer invoked at proof line 994; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Suppose that $\mathcal{G}_i$, $i \in I$ is set of abelian sheaves on $\mathcal{C}$. There exists an ordinal β such that for any sheaf $\mathcal{F}$, any $i \in I$, and any map $\varphi : \mathcal{G}_i \to J_\beta(\mathcal{F})$ there exists an $\alpha < \beta$ such that φ factors through $J_\alpha(\mathcal{F})$.*

Proof. This reduces to the case of a single sheaf $\mathcal{G}$ by taking the direct sum of all the $\mathcal{G}_i$.

Consider the sets

$$S = \coprod\nolimits_{U \in \mathrm{Ob}(\mathcal{C})} \mathcal{G}(U).$$

and

$$T_\beta = \coprod\nolimits_{U \in \mathrm{Ob}(\mathcal{C})} J_\beta(\mathcal{F})(U)$$

The transition maps between the sets T_β are injective. If the cofinality of β is large enough, then $T_\beta = \mathrm{colim}_{\alpha<\beta} T_\alpha$, see Sites, Lemma 0GS0. A morphism $\mathcal{G} \to J_\beta(\mathcal{F})$ factors through $J_\alpha(\mathcal{F})$ if and only if the associated map $S \to T_\beta$ factors through T_α. By Sets, Lemma 05N2 if the cofinality of β is bigger than the cardinality of S, then the result of the lemma is true. Hence the lemma follows from the fact that there are ordinals with arbitrarily large cofinality, see Sets, Proposition 05N3. □